CHEMISTRY MARKING SCHEME
FORM TWO 2026

SECTION A (15 marks)

1 Multiple choices

I	II	III	IV	V	VI	VII	VIII	IX	X
A	C	B	D	B	A	C	D	A	C

(01 mark @ = 10 marks)

2 Matching items

I	II	III	IV	V
D	G	F	C	I

(01 mark @ = 05 marks)

3. (i) In agriculture, some isotopes are used to study how plants absorb nutrients. This enable scientist to improve fertilisers for production of healthier crops.
example phosphorus-32, is used to track how plants take in nutrients from the soil.

(ii) In industrial and environmental fields,
- Tritium-3 track how water moves underground and hence enable scientists to understand sources of drinking water.
- Chlorine isotopes are used to study movement of chlorine in water sources

3 (ii) such as rivers, lakes and underground water.

- Sodium-24 is used to detect leaks in underground pipes.

(iii) In medicine, some isotopes are used in diagnosing and treating diseases.
Example Iodine-131 is used in hospital to diagnose and treating thyroid

(iv) In electricity production, some radioactive isotopes are used in nuclear reactors to generate electricity.
Example Uranium-235.

(v) In Archaeology, archaeologists and historians use Carbon-14 to determine the age of ancient objects.

(02 mark @ = 10 marks)

4. (a)

(i) To wear personal protective gears such as laboratory coat, safety goggles, closed shoes and gloves in order to minimise exposure to hazard.

(ii) To follow safety protocols for handling chemicals and apparatus

(iii.) To ensure proper ventilation

(iv) To know how to treat spills of different chemicals present in the laboratory

(v) To be aware on how to use fire extinguishers and first aid kit items.

(vi) To know the location of all exists and to have easy access to them.

(Only four (4) points, 01 mark @ = 04 marks)

(b) (i) Reasons for adhering to chemistry laboratory safety measures.

(i) In order to prevent accidents and injuries such as cuts, suffocation etc.

(ii) In order to protect equipment~~at~~ from damage such as breakage and corrosion

(b) (iii) In order to ensure accurate experimental results

(iv) In order to comply with laboratory regulations and ethical standards.

(Only three (3) points, 01 @ = 03 marks)

(c) The right way of positioning a victim of shock is by placing him or her in ~~circulating~~ supine position with legs elevated (about 30 cm).

- This position helps to improve venous return to the heart thereby enhancing circulation of blood to the vital organs
- It is also helps to keep the airway open and monitor breathing.

(03 marks
position 01 mark and
advantages 02 marks)

5 (a) (i) Carbon dioxide (01 mark)

(ii) Molecula formula is CO_2 (01 mark)

(iii) Type of bond is Covalent bond (01 mark)

(iv) Any two examples of covalent bonds compounds

- Ammonia (NH_3)
- water (H_2O)
- Methane (CH_4)

Any two (02) examples

(00½ @ = 01 mark)

(b) solution

(i) PbS_2

- $PbS_2 = 0$. (00½ mark)
- $Pb = +4$

$4 + 2S = 0$

$2S = -4$

$\frac{2S}{2} = \frac{-4}{2}$

$S = -2$ (00½ mark)

∴ The oxidation state of sulphur is -2 (00½ marks)

5 (b) (i)

Cu_2SO_4

$-Cu_2SO_4 = 0$

$-SO_4 = -2$ (00½ marks)

Then

$2Cu + -2 = 0$

$2Cu = +2$

$\frac{2Cu}{2} = \frac{+2}{2}$ (00½ marks)

$Cu = +1$

∴ The oxidation state of Cu in Cu_2SO_4 is +1 (00½ mark)

(iii) HCO_3^-

$-HCO_3 = -1$ (00½ marks)

$H = +1$

$O = -2$

Then

$1 + C + -2 \times 3 = -1$

$1 + C + -6 = -1$

$C + -5 = -1$ (00½ mark)

$C = -1 + 5$

$C = 4$

∴ The oxidation state of C is +4 (00½ mark)

5 (b) (iv)

NH_4OH

(00 1/2 mark)

$NH_4OH = 0$

$-NH_4 = +1$

$-H = +1$

$1 + O + 1 = 0$

(00 1/2 mark)

$O + 2 = 0$

$O = -2$

∴ The oxidation state of O_2 is -2 (00 1/2 mark)

6 (a)

SUB ATOMIC PARTICLE

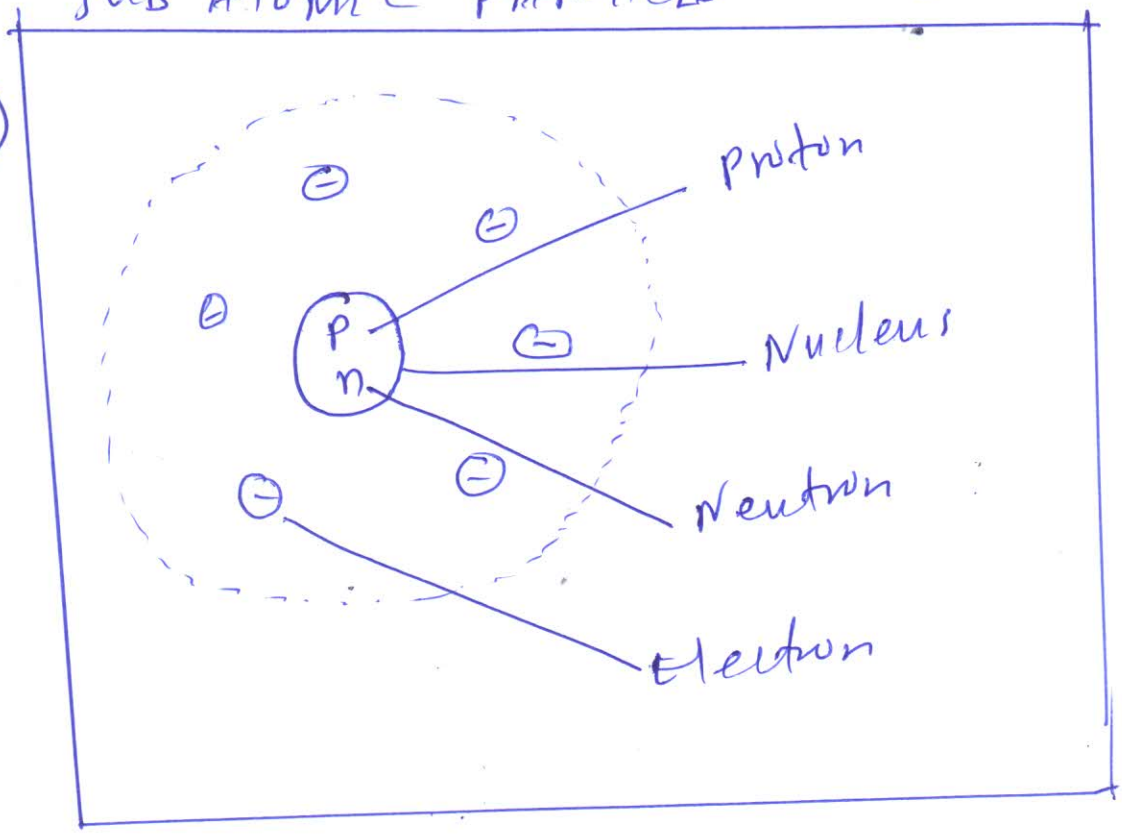


Diagram = 01 mark

heading = ~~00 1/2~~ 01

labelling 00 1/2 @ = 02 marks

(Total marks = ~~03 1/2~~ 04 marks)

(b) (i) $^{27}_{13}Al$

$p = 13$

$n = 14$

$e^- = 13$

(00 ½ @ = 1½ marks)

(ii) $^{137}_{56}Ba$

$p = 56$

$n = 81$

$e^- = 56$

(00 ½ @ = 1½ marks)

(iii) $^{90}_{38}Sr$

$p = 38$

$n = 52$

$e^- = 38$.

(00 ½ @ = 1½ marks)

(iv) $^{235}_{92}U$

$p = 92$

$n = 143$

$e^- = 92$

(00 ½ @ = 1½ marks)

(a) The aluminium pot is better for cooking (00½ mark)

- Reason: Because it is a good conductor of heat compared to china bottle materials which is the poor conductor of heat (02 marks)

(b) (i) They good are good conductor of heat hence used in making cookware example aluminium and iron

(ii) They are good conductor of electricity

(iii) They are ductile and malleable

(iv) They are lustrous

(v) They high boiling and melting point

(vi) They are more reactive

Any other 03 points

(02½ marks @ = 07½ marks)

8 (a) (i) Making thorough observations
(ii) Recording accurately what has been observed
(iii) organising the observed and recorded information
(iv) Repeating test to ensure observations are accurate.
(v) Drawing conclusions from observations
(vi) Predicting possible outcomes of similar experiments.

(Or mark (a) = 08 marks)

(b) (i) Homecare products
These are chemical made materials used to clean home and its surrounding eg soaps, detergents, disinfectants, air freshner and polish.

(ii) Biotechnology
Refers to the use of living organisms to produce economically useful products such as antibiotics and hormones by using skills from Chemistry, biology and chemical engineering

(iii) Agro chemistry - Is the scientific field which examines the chemical processes related to plants, animals and soil health, crop ~~protection~~ protections and fertilizers

(b) (iii) in order to ensure good availability and security as well as good agricultural practices and quality of agricultural products

(iv) Geo-chemists

Refers to the people who apply chemistry principles and theories in exploring the chemical compositions and processes of earth and other planets

(01 mark @ = 04 marks)

(c) A - Melting

B - vaporization / Evaporation

C - Ionisation

D - Deionisation

E - Condensation

F - Freezing

G - Deposition

H - Sublimation

(01 @ = 08 marks)

9 (b).

- The change of state of matter especially evaporation and condensation are used in simple distillation and fractional distillation to separate pure water from impurities
- Example, mixture of salt and water (Evaporation), muddy water (simple distillation) and mixture of ethanol and water (fractional distillation)

(02 marks).

10 The followings are procedures to someone who has got electric shock

Procedure

1. Break the contact between the electrical source and the victim by switching off the power source.
2. If it is not possible to switch off the current, move the person from his or her position by a dry non-conducting material made of cardboard, plastic or wooden plank or wood.

3. If there is an immediate danger such as severe burns, muscle pain, difficulty breathing and confusion call for a professional help immediately.

4. If the victim is unconsious, check the pulse rate and whether the victim is breathing. If he or she has breathing problems, try to revive his or her breathing rate to normal.

5. Administer first Aid for burns, shock or other injuries the victim may have sustained

6. Take the victim to a nearby health care facility

(02½@ = 15 marks)